

Nested Harmonic Jets and Gamma Renormalization

Exact Stieltjes Identities, Regulator Conversion,
Cyclotomic Products, and All-Depth Polylogarithmic Resonance

Research continuation prepared for Vladimir Reshetnikov
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Abstract

A single finite harmonic product connects three different operations: taking a nested-sum generating function, removing a spectral pole, and forming an Abel transform. We develop their exact relation, keeping their normalizations separate. A holomorphic two-variable renormalization has the Gamma quotient $\Gamma(a)/\Gamma(a+z)$ as its spectral value at one. Every spectral jet is then expressed by a Bell polynomial in generalized Stieltjes constants and Hurwitz-zeta derivatives. This gives a finite all-depth conversion from harmonic regularization to diagonal Laurent finite parts, including explicit correction terms through depth four. Parameter multiplication has a necessary exponential correction, and a higher-genus cyclotomic identity reduces fractional-order products to the same Gamma quotient.

For equal-order multiple polylogarithms we classify all depth weights at which the complete depth-generating function becomes a polynomial in its argument. At a negative integral depth weight, its first fixed-weight spectral derivative is elementary. Higher derivatives have explicit logarithmic harmonic Bell coefficients and reduce to shifted logarithmic Euler sums of nesting depth at most the derivative order. These results include ordinary and Euler antiderivative formulas and convergent endpoint identities. The proofs are analytic and algebraic; 826 exact finite assertions and 1,075 high-precision numerical comparisons accompany them. The latter are diagnostics, not interval certificates. Classical symmetric-function, Gamma-product, and regularization results are credited separately; global priority for the derived families is not asserted.

Scope. This is an identity-focused continuation, not a claim to settle the remaining S_6 or S_8 special-value problems or numerical period independence. The repository was read at the pinned revision above. No repository files were modified and no proof-assistant formalization is claimed.

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1 Research question, sources, and contribution

The integration and differentiation chapters of the ProveIt polylogarithm manuscript organize single Hurwitz-zeta jets, generalized Stieltjes functions, and harmonic-number coefficient laws. Its depth chapter and research programme ask for exact reductions rather than extrapolations from an integer-relation search. The present continuation asks a compatible but different question:

Can the spectral derivatives of *all nested depths at once* be controlled by one analytic object, and can special depth weights force exact reductions?

The answer is affirmative for the equal-order family studied here. There are two complementary reductions. At the singular endpoint in the spectral variable, a Gamma-normalized product gives a finite Stieltjes–zeta coefficient calculus. In the open argument disk, a zero of one finite-product factor collapses an infinite depth-generating function to a polynomial. Differentiating near that zero gives the elementary first-order identities and the higher-order bounded-nesting theorem.

1.1 What is classical and what is established here

The finite elementary-symmetric-product identity, Newton’s identities, the Weierstrass product for Gamma, and the relation between symmetric functions and multiple zeta values are classical. Hoffman develops the latter connection, including the role of the Gamma genus [6, 7]. Regularization and double-shuffle structures have a substantial independent theory [8, 9]. We do not identify every regularization with every other one, nor call the reciprocal-Gamma generating function a new special-function evaluation.

The results proved in this article are the following coordinated statements. Theorem 3.1 proves a locally uniform cutoff realization of the spectral family. Theorems 4.1 and 5.1 give all-order jet and regulator-conversion formulas with precise finite coefficient requirements. Theorems 6.1 and 7.2 identify the parameter-multiplication correction and the cyclotomic fractional-order Gamma reduction. Theorems 10.1, 10.2, and 11.1 classify polynomial truncations, evaluate every first resonant derivative elementarily, and give every higher resonant coefficient. Corollary 11.3 specifies the comparison class in the resulting bounded-nesting reduction. Their derivations are included in full. An exhaustive literature priority investigation is not part of this delivery.

1.2 Pinned repository and targeted reading

The source revision is

fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed.

The main source areas are `Analysis/Polylogarithms/docs/manuscript` and `docs/incoming` [1]. The main manuscript, relevant portions of the depth, integration, differentiation, and discovery chapters, and the incoming inventory/intake instructions were inspected. The filenames and README summaries of all five incoming archives at this snapshot were also examined. Their complete manuscripts and verification outputs were *not* exhaustively audited.

Those incoming reports already develop shifted Stieltjes products, convolution closure, contact terms, ordinary log-Gamma correlations, and related Lerch limits. The present work does not repackage those bilinear identities as new results. Its word “resonance” refers to the zero $1 - mk^{-s} = 0$ in a *depth-generating product*, not to an integer-order Lerch expansion. The source audit in Section 12 states the remaining limitations and a specific editorial correction.

2 Finite harmonic products and conventions

Fix a real parameter $a > 0$, and put $x_n = n + a$. All powers x_n^s use the real logarithm of the positive base. Spectral differentiation always means d/ds , not differentiation in a . Define

$$H_N^{(s)}(a) = \sum_{n=0}^{N-1} x_n^{-s}, \quad (2.1)$$

$$E_{d,N}(s; a) = \sum_{0 \leq n_1 < \dots < n_d < N} (x_{n_1} \cdots x_{n_d})^{-s}, \quad E_{0,N} = 1. \quad (2.2)$$

The function $E_{d,N}$ vanishes for $d > N$. Equal orders make increasing or decreasing index notation equivalent; the multiple-polylogarithm notation introduced later will use decreasing indices.

Proposition 2.1 (Finite product and Newton recurrence). *For every N ,*

$$P_N(s, z; a) := \prod_{n=0}^{N-1} (1 + zx_n^{-s}) = \sum_{d=0}^N z^d E_{d,N}(s; a), \quad (2.3)$$

and

$$dE_{d,N}(s; a) = \sum_{k=1}^d (-1)^{k-1} H_N^{(ks)}(a) E_{d-k,N}(s; a). \quad (2.4)$$

Proof. Choose the term zx_n^{-s} from exactly d factors of the product to obtain (2.3). As a formal power series in z ,

$$z \partial_z \log P_N = \sum_{k \geq 1} (-1)^{k-1} H_N^{(ks)}(a) z^k.$$

Multiplication by P_N and comparison of coefficients give (2.4). Only finite polynomials are needed for each coefficient. $\square$

Thus generalized harmonic numbers are the power-sum coordinates of the nested family. For example,

$$E_{2,N} = \frac{(H_N^{(s)})^2 - H_N^{(2s)}}{2}, \quad E_{3,N} = \frac{(H_N^{(s)})^3 - 3H_N^{(s)}H_N^{(2s)} + 2H_N^{(3s)}}{6}.$$

Their spectral derivatives insert logarithms:

$$\partial_s^r H_N^{(ks)}(a) = (-k)^r \sum_{n < N} \frac{\log^r x_n}{x_n^{ks}}. \quad (2.5)$$

The factor k^r in this equation is essential throughout the paper.

We use the Laurent convention

$$\zeta(1 + \varepsilon, a) = \frac{1}{\varepsilon} + \sum_{r \geq 0} \frac{(-1)^r \gamma_r(a)}{r!} \varepsilon^r, \quad \gamma_0(a) = -\psi(a). \quad (2.6)$$

The standard Hurwitz identities and their domains are recorded in DLMF §25.11 [4]. In particular $\partial_a \zeta(s, a) = -s\zeta(s+1, a)$ and $\zeta(k, a) = (-1)^k \psi^{(k-1)}(a)/(k-1)!$ for $k \geq 2$.

The complete exponential Bell polynomials Y_r are defined by

$$\exp \left(\sum_{j \geq 1} X_j \frac{t^j}{j!} \right) = \sum_{r \geq 0} Y_r(X_1, \dots, X_r) \frac{t^r}{r!}. \quad (2.7)$$

Thus $Y_0 = 1$, $Y_1 = X_1$, $Y_2 = X_1^2 + X_2$, and $Y_3 = X_1^3 + 3X_1X_2 + X_3$.

3 A holomorphic cutoff renormalization

Set $X = N + a$ and define the entire-in- s function

$$C_N(s; a) = \frac{1 - X^{1-s}}{s - 1}, \quad C_N(1; a) = \log X. \quad (3.1)$$

The endpoint $N + a$, and not a silently substituted endpoint, is part of the normalization. Its derivatives at one are

$$\partial_s^r C_N(1; a) = \frac{(-1)^r}{r + 1} \log^{r+1} X. \quad (3.2)$$

Define

$$\mathcal{G}_N(s, z; a) = e^{-z C_N(s; a)} P_N(s, z; a). \quad (3.3)$$

Theorem 3.1 (Cutoff limit and the Gamma slice). *For fixed $a > 0$, the functions $\mathcal{G}_N$ converge locally uniformly on $\{\operatorname{Re} s > 1/2\} \times \mathbb{C}$ to a holomorphic function $\mathcal{G}$, entire in z , given by*

$$\mathcal{G}(s, z; a) = e^{z(\zeta(s, a) - 1/(s-1))} \prod_{n \geq 0} (1 + z x_n^{-s}) e^{-z x_n^{-s}}. \quad (3.4)$$

The apparent singularity at $s = 1$ is removable. All fixed mixed s, z derivatives converge locally uniformly. Moreover

$$\boxed{\mathcal{G}(1, z; a) = \frac{\Gamma(a)}{\Gamma(a + z)}}. \quad (3.5)$$

The zeros of $\mathcal{G}(s, \cdot; a)$ are exactly $z = -x_n^s$, each simple.

Proof. Euler summation gives, locally uniformly for $\operatorname{Re} s > 0$,

$$\zeta(s, X) - \frac{X^{1-s}}{s - 1} = O(X^{-\operatorname{Re} s}),$$

where the difference is interpreted at one by removability. One direct justification is to subtract the integral of $(X + t)^{-s}$ from its sum over nonnegative integers, interval by interval. The resulting series of differences is bounded on a compact s -set by a constant times $\int_0^\infty (X + t)^{-\operatorname{Re} s - 1} dt$, plus its endpoint term. Since $H_N^{(s)}(a) = \zeta(s, a) - \zeta(s, X)$, this proves

$$H_N^{(s)}(a) - C_N(s; a) \longrightarrow \zeta(s, a) - \frac{1}{s - 1}.$$

For a compact set in the stated domain, choose $\sigma > 1/2$ below every real part and a uniform bound for $|z|$. The tail factors satisfy

$$\log(1 + z x_n^{-s}) - z x_n^{-s} = O(x_n^{-2\sigma})$$

for all sufficiently large n , with the logarithm chosen near one. The summable majorant proves normal convergence of the centered product. Combining its convergence with that of the linear exponential proves (3.4). The holomorphic convergence theorem and Cauchy's integral formula give every fixed derivative assertion.

At one the first exponential is $e^{-z\psi(a)}$. The Weierstrass Gamma product [2] gives

$$\frac{\Gamma(a)}{\Gamma(a + z)} = e^{-z\psi(a)} \prod_{n \geq 0} \left(1 + \frac{z}{x_n}\right) e^{-z/x_n},$$

proving (3.5). The exponential never vanishes. Away from the displayed linear-factor zeros, the normally convergent tail has a nonzero logarithm and hence is nonzero. Distinct x_n give distinct moduli $x_n^{\text{Re } s}$, so the factors have no coincident zeros. $\square$

Corollary 3.2 (Harmonic regularized depth coefficients). *Write $g_d(a) = [z^d]\Gamma(a)/\Gamma(a+z)$. Then*

$$g_d(a) = \lim_{N \rightarrow \infty} \sum_{j=0}^d \frac{(-\log(N+a))^j}{j!} E_{d-j,N}(1; a). \quad (3.6)$$

More generally, for every fixed d, r , coefficient extraction and r spectral derivatives commute with the limit in (3.3).

Proof. Expand the finite exponential in z , then apply the local uniform convergence on a circle about $z = 0$. Apply Cauchy's formula also in s for the derivative statement. $\square$

This is a coefficientwise regularization, not an ordinary value assigned to a divergent nested sum without a subtraction prescription.

4 The Stieltjes–zeta spectral jet calculus

For z initially near zero define the logarithm by $\log \mathcal{G}(s, 0; a) = 0$, and put

$$\Lambda_r(z; a) = \partial_s^r \log \mathcal{G}(s, z; a)|_{s=1}, \quad F_r(z; a) = \partial_s^r \mathcal{G}(s, z; a)|_{s=1}. \quad (4.1)$$

Theorem 4.1 (All spectral jets). *For $r \geq 0$ and $|z| < a$,*

$$\boxed{\Lambda_r(z; a) = (-1)^r z \gamma_r(a) + \sum_{k \geq 2} (-1)^{k-1} k^{r-1} z^k \zeta^{(r)}(k, a)}. \quad (4.2)$$

Here $\zeta^{(r)}$ differentiates its first variable and $k^{r-1} = 1/k$ when $r = 0$. For $r \geq 1$,

$$F_r(z; a) = \frac{\Gamma(a)}{\Gamma(a+z)} Y_r(\Lambda_1(z; a), \dots, \Lambda_r(z; a)). \quad (4.3)$$

Each F_r is entire. The apparent poles on the right of (4.3) cancel. The functions Λ_r , $r \geq 1$, continue meromorphically to the z -plane, with possible poles only at $-a, -a-1, \dots$

Proof. For small z the logarithm of (3.4) is

$$\log \mathcal{G}(s, z; a) = z \left(\zeta(s, a) - \frac{1}{s-1} \right) + \sum_{k \geq 2} \frac{(-1)^{k-1} z^k}{k} \zeta(ks, a). \quad (4.4)$$

Normal convergence in a neighborhood of $s = 1$ allows spectral differentiation. Equation (2.6) gives the first term, while $d^r \zeta(ks, a)/ds^r = k^r \zeta^{(r)}(ks, a)$ gives the factor in (4.2). Expanding the exponential in the increment $s - 1$ proves (4.3). Entireness of F_r follows from Theorem 3.1, independently of any cancellation manipulation.

For continuation of Λ_r , separate any finite number of factors from the product. A spectral derivative of their logarithms is rational in z for $r \geq 1$. The remaining differentiated centered logarithmic series converges normally away from its factor zeros: a fixed compact set has a summable tail of size $O(x_n^{-2} \log^r x_n)$. This proves the meromorphic statement and also the cancellation claim in (4.3). $\square$

At order zero, $\Lambda_0 = \log \Gamma(a) - \log \Gamma(a + z)$ denotes the analytic logarithm continued from zero, not the principal logarithm of a possibly winding Gamma value. The first depth coefficients are

$$g_1 = c, \quad g_2 = \frac{c^2 - Z_2}{2}, \quad (4.5)$$

$$g_3 = \frac{c^3 - 3cZ_2 + 2Z_3}{6}, \quad (4.6)$$

$$g_4 = \frac{c^4 - 6c^2Z_2 + 3Z_2^2 + 8cZ_3 - 6Z_4}{24}, \quad (4.7)$$

where $c = \gamma_0(a)$ and $Z_k = \zeta(k, a)$. At zero spectral order, the Z_k may equally be replaced by ordinary polygamma values.

4.1 An explicit logarithmic-harmonic limit

Differentiating (3.3) logarithmically before taking the limit gives

$$\Lambda_1(z; a) = \lim_{N \rightarrow \infty} \left\{ \frac{z}{2} \log^2(N + a) - \sum_{n < N} \frac{z \log x_n}{x_n + z} \right\}. \quad (4.8)$$

This identity holds away from the poles and locally uniformly there. It expresses a Stieltjes–zeta jet as a renormalized generalized harmonic sum with a shifted denominator. Higher limits follow from (3.2); they must differentiate the counterterm too.

4.2 Exact cancellation at a moving Gamma zero

Proposition 4.2 (Zero-flow identities). *Let $x = a + m$ with $m \geq 0$. Then*

$$F_1(-x; a) = x \log x \Gamma(a) (-1)^m m!. \quad (4.9)$$

More generally, with $V_x = \partial_s + (\log x)z\partial_z$,

$$(V_x^r \mathcal{G})(s, -x^s; a) = 0 \quad (r \geq 0, \operatorname{Re} s > 1/2). \quad (4.10)$$

Proof. The path $z(s) = -x^s$ is a zero of the corresponding linear factor of (3.4). Its total derivative is precisely V_x , proving (4.10). At one the $r = 1$ case says $F_1(-x) = x \log x \partial_z \mathcal{G}(1, -x)$. The reciprocal Gamma function has derivative $(-1)^m m!$ at $-m$; (3.5) gives (4.9). $\square$

Thus a zero of the Gamma slice need not be a zero of its spectral derivative. For instance $x = 1$ gives zero because $\log x = 0$, whereas a generic x gives the nonzero value in (4.9).

5 Diagonal Laurent finite parts are a different regulator

For $\operatorname{Re} s > 1$ define the diagonal multiple Hurwitz zeta value

$$Z_d(s; a) = \sum_{0 \leq n_1 < \dots < n_d} (x_{n_1} \cdots x_{n_d})^{-s}, \quad Z_0 = 1. \quad (5.1)$$

Newton's identities express every fixed Z_d as a polynomial in $\zeta(s, a), \dots, \zeta(ds, a)$, so it is meromorphic in s . Its Laurent finite part at one means the coefficient of $(s - 1)^0$ of *this specific one-variable restriction*.

Theorem 5.1 (Exact conversion of the two regulators). *As a formal depth series with meromorphic coefficients near $\varepsilon = 0$,*

$$\sum_{d \geq 0} z^d Z_d(1 + \varepsilon; a) = e^{z/\varepsilon} \mathcal{G}(1 + \varepsilon, z; a). \quad (5.2)$$

Consequently, for $d \geq 0$ and any integer v ,

$$[\varepsilon^v] Z_d(1 + \varepsilon; a) = \sum_{j=0}^d \frac{[z^{d-j} \varepsilon^{v+j}] \mathcal{G}(1 + \varepsilon, z; a)}{j!}, \quad (5.3)$$

where a negative Taylor index on $\mathcal{G}$ means zero. In particular,

$$\boxed{\text{FP}_{s=1} Z_d(s; a) = \sum_{j=0}^d \frac{[z^{d-j}] F_j(z; a)}{(j!)^2}}. \quad (5.4)$$

The finite part belongs to the finite polynomial algebra over $\mathbb{Q}$ generated by

$$\gamma_0(a), \dots, \gamma_{d-1}(a), \quad \zeta^{(j)}(k, a) \quad (2 \leq k \leq d, \ 0 \leq j \leq d - k) \quad (5.5)$$

for $d \geq 1$.

Proof. In the absolutely convergent region the depth-generating function of (5.1) is the uncentered infinite product. Equation (3.4) shows that it equals $e^{z/(s-1)} \mathcal{G}(s, z; a)$. Alternatively, logarithmic expansion gives the same identity immediately from Newton's recurrence. For each fixed depth it is an equality of meromorphic functions, so it continues to a punctured neighborhood of one.

In a fixed coefficient z^d , the exponential contributes only its terms $j = 0, \dots, d$. Extracting the Laurent coefficient proves (5.3). Since $[\varepsilon^j] \mathcal{G} = F_j/j!$, this gives (5.4), including the second factorial. The atom restriction follows by substituting (4.2): a z^k zeta term needs k units of depth, leaving at most $d - k$ spectral derivatives from the pole factor. A linear Stieltjes term leaves at most $d - 1$ derivatives. Products of such terms add no new generators. $\square$

The first nontrivial formulas, with $c = \gamma_0(a)$, are

$$\text{FP } Z_2 = g_2 - \gamma_1(a), \quad (5.6)$$

$$\text{FP } Z_3 = g_3 - c\gamma_1(a) - \zeta'(2, a) + \frac{\gamma_2(a)}{4}, \quad (5.7)$$

$$\begin{aligned} \text{FP } Z_4 = g_4 + \frac{\zeta(2, a)\gamma_1(a)}{2} - c\zeta'(2, a) - \frac{\zeta''(2, a)}{2} + \zeta'(3, a) \\ - \frac{c^2\gamma_1(a)}{2} + \frac{c\gamma_2(a)}{4} + \frac{\gamma_1(a)^2}{4} - \frac{\gamma_3(a)}{36}. \end{aligned} \quad (5.8)$$

For example, the elementary diagonal identity $Z_2(s; a) = (\zeta(s, a)^2 - \zeta(2s, a))/2$ already proves (5.6). Its extra $-\gamma_1(a)$ comes from multiplying the spectral pole by the next Laurent coefficient. Setting $s = 1$ after removing only the depth-one pole does not reproduce this finite part.

Remark 5.2 (Scope of “regularized”). Harmonic cutoff regularization (3.6), the diagonal Laurent finite part (5.4), and shuffle regularization are not interchangeable names. The identities above specify the first two directly. They do not assert that raw diagonal finite-part extraction is a shuffle character. The general double-shuffle regularization theory [8] addresses additional algebraic compatibility questions. A different finite subtraction constant changes the harmonic normalization by an exponential in z .

6 Parameter multiplication and its spectral correction

Theorem 6.1 (Shift and multiplication identities). *For $a > 0$, $\operatorname{Re} s > 1/2$, and all z ,*

$$\mathcal{G}(s, z; a) = (1 + z/a^s) \mathcal{G}(s, z; a + 1). \quad (6.1)$$

For every integer $q \geq 2$,

$$\prod_{j=0}^{q-1} \mathcal{G}\left(s, z; \frac{a+j}{q}\right) = \exp\left(z \frac{q^s - q}{s-1}\right) \mathcal{G}(s, q^s z; a). \quad (6.2)$$

The quotient in the exponential is removable at one, where its value is $q \log q$.

Proof. The shift removes the $n = 0$ term from the uncentered product; the centered exponential correction removes exactly the same term, giving (6.1). For multiplication, use

$$\sum_{j=0}^{q-1} \zeta\left(t, \frac{a+j}{q}\right) = q^t \zeta(t, a),$$

Insert this identity into (4.4). Terms of degree $k \geq 2$ agree with those of $\log \mathcal{G}(s, q^s z; a)$. The difference in degree one is $z(q^s - q)/(s-1)$. Exponentiate first near $z = 0$ and extend by entireness. The same proof may be carried out in the absolutely convergent region before analytic continuation. $\square$

At $s = 1$, (6.2) is the Gamma multiplication ratio

$$\prod_{j < q} \frac{\Gamma((a+j)/q)}{\Gamma((a+j)/q + z)} = q^{qz} \frac{\Gamma(a)}{\Gamma(a + qz)}. \quad (6.3)$$

The classical Gamma multiplication formula is recorded in DLMF §5.5 [3]. Omitting the exponential in (6.2) would already contradict this specialization.

Corollary 6.2 (All-order distribution of logarithmic jets). *Let $\ell = \log q$ and $D = w\partial_w$. Then, as germs at $z = 0$,*

$$\begin{aligned} \sum_{j=0}^{q-1} \Lambda_r\left(z; \frac{a+j}{q}\right) &= \frac{qz\ell^{r+1}}{r+1} \\ &+ \sum_{h=0}^r \binom{r}{h} \ell^{r-h} D^{r-h} \Lambda_h(w; a) \Big|_{w=qz}. \end{aligned} \quad (6.4)$$

For $r = 1$ this becomes

$$\sum_{j < q} \Lambda_1\left(z; \frac{a+j}{q}\right) = \Lambda_1(qz; a) - qz\ell \psi(a + qz) + \frac{qz\ell^2}{2}. \quad (6.5)$$

Proof. The r -th derivative at one of $(q^s - q)/(s-1)$ is $q\ell^{r+1}/(r+1)$. Along $w = q^s z$, differentiation of a function of (s, w) acts as $\partial_s + \ell D$; the two operators commute. Expanding its r -th power gives (6.4). Finally $D\Lambda_0(w; a) = -w\psi(a + w)$ gives (6.5). $\square$

This identity links Stieltjes jets, spectral zeta derivatives, and a polygamma argument shift without any numerical relation search. Further z derivatives of (6.5) produce explicit higher polygamma terms and remain valid meromorphically away from the displayed poles.

7 Higher-genus products and a cyclotomic Gamma identity

The half-plane in Theorem 3.1 is tied to a quadratic tail. To cross it, higher power sums must also be subtracted. This can be done without introducing an unspecified regularization.

Definition 7.1. For an integer $M \geq 1$ define

$$\mathcal{G}_N^{[M]}(s, z; a) = P_N(s, z; a) \exp \left(- \sum_{k=1}^M \frac{(-1)^{k-1} z^k}{k} C_N(ks; a) \right). \quad (7.1)$$

Thus $\mathcal{G}_N^{[1]} = \mathcal{G}_N$.

Theorem 7.2 (Higher-genus limit and cyclotomic reduction). *The functions in (7.1) converge locally uniformly for $\operatorname{Re} s > 1/(M+1)$ and all z , to*

$$\begin{aligned} \mathcal{G}^{[M]}(s, z; a) = & \exp \left(\sum_{k=1}^M \frac{(-1)^{k-1} z^k}{k} \left(\zeta(ks, a) - \frac{1}{ks-1} \right) \right) \\ & \times \prod_{n \geq 0} (1 + zx_n^{-s}) \exp \left(\sum_{k=1}^M \frac{(-1)^k z^k}{kx_n^{ks}} \right). \end{aligned} \quad (7.2)$$

Every apparent pole at $s = 1/k$, $1 \leq k \leq M$, is removed in the specified combination. Put $\omega = e^{2\pi i/m}$ and $u = (-1)^{m+1} z^m$. For $m \geq 1$ and $\operatorname{Re} s > 1/(m+1)$,

$$\prod_{j=0}^{m-1} \mathcal{G}^{[m]}(s, \omega^j z; a) = \mathcal{G}(ms, u; a). \quad (7.3)$$

In particular,

$$\prod_{j=0}^{m-1} \mathcal{G}^{[m]}(1/m, \omega^j z; a) = \frac{\Gamma(a)}{\Gamma(a + (-1)^{m+1} z^m)}. \quad (7.4)$$

For logarithms continued from $z = 0$, every $r \geq 0$ also satisfies

$$\sum_{j=0}^{m-1} \partial_s^r \log \mathcal{G}^{[m]}(s, \omega^j z; a) \Big|_{s=1/m} = m^r \Lambda_r((-1)^{m+1} z^m; a). \quad (7.5)$$

Proof. Separate the first M terms of the logarithmic expansion of each factor. The remaining tail is $O(x_n^{-(M+1)\sigma})$ on a compact set with $\operatorname{Re} s \geq \sigma > 1/(M+1)$. For each $k \leq M$, the Euler-summation argument in Theorem 3.1 applies to ks , and proves convergence of $H_N^{(ks)} - C_N(ks)$ to its pole-subtracted zeta value. This proves (7.2), holomorphy, and convergence of all fixed derivatives.

The cyclotomic assertion is already an exact finite-cutoff identity. Indeed

$$\prod_{j=0}^{m-1} (1 + \omega^j t) = 1 - (-t)^m = 1 + (-1)^{m+1} t^m,$$

and $\sum_{j < m} \omega^{jk} = 0$ unless m divides k . In the counterterm sum with $1 \leq k \leq m$, only $k = m$ remains; it is precisely $-uC_N(ms; a)$. Hence

$$\prod_{j < m} \mathcal{G}_N^{[m]}(s, \omega^j z; a) = \mathcal{G}_N(ms, u; a).$$

Both sides have locally uniform limits in the stated half-plane, since $m \operatorname{Re} s > m/(m+1) \geq 1/2$. Taking the limit proves (7.3). The Gamma slice gives (7.4). Logarithmic differentiation of (7.3), on a neighborhood free of zeros, gives (7.5); each spectral derivative contributes a factor m . $\square$

For example the half-order pair has the exact identity

$$\mathcal{G}^{[2]}(1/2, z; a) \mathcal{G}^{[2]}(1/2, -z; a) = \frac{\Gamma(a)}{\Gamma(a - z^2)}. \quad (7.6)$$

At third order the corresponding three rotated factors give $\Gamma(a)/\Gamma(a + z^3)$. Their individual jets involve pole-subtracted Hurwitz values at fractional spectral arguments; the cyclotomic sum removes all components except the ordinary Stieltjes–zeta jet on the right of (7.5). No choice of a branch of $z^{1/m}$ is needed.

8 Polylogarithmic Euler primitives and ordinary antiderivatives

The spectral coefficients admit a second summation direction. This turns Euler integration in the depth variable into a shift in an ordinary polylogarithm’s order.

Theorem 8.1 (A complete normalized primitive ladder). *For integers $r, q \geq 0$, initially for $|z| < a$, put*

$$A_{r,q}(z; a) = (-1)^r z \gamma_r(a) + \sum_{k \geq 2} (-1)^{k-1} k^{r-q-1} z^k \zeta^{(r)}(k, a). \quad (8.1)$$

Then $A_{r,0} = \Lambda_r$, $A_{r,q}(0; a) = 0$, and

$$z \partial_z A_{r,q+1}(z; a) = A_{r,q}(z; a). \quad (8.2)$$

An exact normally convergent continuation to $\Omega_a = \mathbb{C} \setminus (-\infty, -a]$ is

$$A_{r,q}(z; a) = (-1)^r \left[z \gamma_r(a) - \sum_{n \geq 0} \log^r x_n \left(\operatorname{Li}_{q+1-r}(-z/x_n) + \frac{z}{x_n} \right) \right]. \quad (8.3)$$

For $r = 0$ the factor $\log^0 x_n$ is defined as one, including $x_n = 1$. The polylogarithms use their principal continuation on their usual slit plane.

Proof. Apply $z \partial_z$ termwise in (8.1), multiplying a degree k term by k . This proves (8.2). For $k \geq 2$ the absolutely convergent Dirichlet representation gives

$$\zeta^{(r)}(k, a) = (-1)^r \sum_{n \geq 0} \frac{\log^r x_n}{x_n^k}.$$

Substitute it into (8.1) and sum first over k using the polylogarithm series [5]. The removed linear term is exactly z/x_n , and the result is (8.3).

For any compact subset of Ω_a , sufficiently large x_n make the bracket $O(|z|^2/x_n^2)$, uniformly on that compact set. Its product with $\log^r x_n$ is summable. The finitely many remaining polylogarithms are holomorphic there; their individual cuts start at $-x_n$ and are contained in the excluded ray. This proves normal convergence and analytic continuation. Equation (8.2) consequently continues too. $\square$

This is a normalized ladder, rather than an ambiguous notation for negative polygamma. Every Euler primitive vanishes at zero. Negative integer orders of the polylogarithm in (8.3) are rational functions. In particular,

$$\Lambda_1(z; a) = -z\gamma_1(a) + z^2 \sum_{n \geq 0} \frac{\log x_n}{x_n(x_n + z)}, \quad (8.4)$$

$$\Lambda_2(z; a) = z\gamma_2(a) - z^2 \sum_{n \geq 0} \frac{(\log x_n)^2(2x_n + z)}{x_n(x_n + z)^2}. \quad (8.5)$$

These formulas also give the meromorphic continuation asserted in Theorem 4.1 for the first two orders.

Corollary 8.2 (Two ordinary antiderivatives). *For $z \in \Omega_a$, with path integration from zero within that domain,*

$$\begin{aligned} \int_0^z \Lambda_1(u; a) du = & -\frac{\gamma_1(a)z^2}{2} \\ & + \sum_{n \geq 0} \log x_n \left[\frac{z^2}{2x_n} - z + x_n \operatorname{Log} \left(1 + \frac{z}{x_n} \right) \right], \end{aligned} \quad (8.6)$$

$$\begin{aligned} \int_0^z \Lambda_0(u; a) du = & z \log \Gamma(a) - \zeta'(-1, a + z) + \zeta'(-1, a) \\ & + az + \frac{z^2 - z}{2} - \frac{z}{2} \log(2\pi). \end{aligned} \quad (8.7)$$

The series in (8.6) is normally convergent.

Proof. The bracket in (8.6) is the integral of $u^2/(x_n(x_n + u))$, and its tail starts with $z^3/(3x_n^2)$. This justifies integration term by term in (8.4). For (8.7), differentiate $\partial_a \zeta(s, a) = -s\zeta(s + 1, a)$ in s at $s = -1$ and use $\zeta'(0, a) = \log \Gamma(a) - \frac{1}{2} \log(2\pi)$ and $\zeta(0, a) = \frac{1}{2} - a$ [4]. The derivative of the asserted right side is $\Lambda_0(z; a)$ and its value at zero is zero. $\square$

Formula (8.7) is the classical log-Gamma primitive in the present normalization, included as a compatibility check rather than a new evaluation. Formula (8.6) is its spectral-jet companion.

9 The complete depth-generating polylogarithm

For $|w| < 1$ use decreasing indices and define

$$\operatorname{Li}_{\{s\}^d}(w, 1, \dots, 1) = \sum_{n_1 > \dots > n_d \geq 1} \frac{w^{n_1}}{(n_1 \dots n_d)^s}. \quad (9.1)$$

In this notation ∂_s differentiates *all d orders simultaneously along the diagonal*. It is not a derivative of the first order with the other orders fixed.

Theorem 9.1 (Depth product, recurrence, and Abel transform). *For $\operatorname{Re} s > 0$, $z \in \mathbb{C}$, and $|w| < 1$,*

$$\mathfrak{F}(s, z, w) = 1 + \sum_{d \geq 1} z^d \operatorname{Li}_{\{s\}^d}(w, 1, \dots, 1) = 1 + \sum_{n \geq 1} b_n(s, z) w^n, \quad (9.2)$$

$$b_n(s, z) = \frac{z}{n^s} \prod_{k=1}^{n-1} \left(1 + \frac{z}{k^s} \right). \quad (9.3)$$

The depth and coefficient series converge absolutely and locally uniformly in this domain, as do all fixed parameter derivatives. With $b_0 = 1$,

$$n^s b_n = ((n-1)^s + z) b_{n-1}, \quad n \geq 1, \quad (9.4)$$

where $0^s = 0$. There is also the exact Abel-transform identity

$$\mathfrak{F}(s, z, w) = (1-w) \sum_{N \geq 0} P_N(s, z; 1) w^N. \quad (9.5)$$

On the slice $s = 1$,

$$\mathfrak{F}(1, z, w) = (1-w)^{-z}. \quad (9.6)$$

Proof. Fix the largest index $n_1 = n$. The remaining factors enumerate elementary symmetric functions of $1^{-s}, \dots, (n-1)^{-s}$. Summing their depths proves (9.3). On a compact set let $\sigma > 0$ bound the real parts from below, $|z| \leq R$, and $|w| \leq \rho < 1$. The absolute contribution at largest index n is at most

$$\rho^n R n^{-\sigma} \exp \left(R \sum_{k < n} k^{-\sigma} \right).$$

The exponent is $o(n)$ for every $\sigma > 0$, so this is summable. It bounds the depth series as well as the coefficient series and proves joint local uniform convergence. Holomorphy and Cauchy's formula then justify all fixed derivatives and regroupings.

Cancel the common product factors to obtain (9.4), or verify it without division by multiplying the products. Since $b_n = P_n(s, z; 1) - P_{n-1}(s, z; 1)$, summation by parts proves (9.5). At $s = 1$, the recurrence is that of the binomial expansion: $b_n = (z)_n / n!$, proving (9.6). $\square$

For $\operatorname{Re} s > 1$ the endpoint series converges absolutely and gives

$$\mathcal{G}(s, z; 1) = e^{-z/(s-1)} \mathfrak{F}(s, z, 1). \quad (9.7)$$

Thus the two principal objects of this article arise from the same finite products; they are not unrelated special-function constructions. Nevertheless their normalizations at $s = 1$ differ:

$$\lim_{N \rightarrow \infty} (N+1)^{-z} P_N(1, z; 1) = \frac{1}{\Gamma(1+z)}, \quad (1-w)^z \mathfrak{F}(1, z, w) = 1. \quad (9.8)$$

The Gamma factor is the exact distinction between this harmonic cutoff and this Abel normalization. One may not replace one by the other without it.

Remark 9.2 (Why the open spectral half-plane matters). At $s = 0$, direct geometric summation, in its initial convergence region, gives

$$\mathfrak{F}(0, z, w) = \frac{1-w}{1-(1+z)w}. \quad (9.9)$$

For fixed nonzero w this has a pole in z . Thus the entireness-in-depth claim of Theorem 9.1 cannot simply be extended to $s = 0$.

For later use define $\theta = w \partial_w$, and define θ^s on a power series by multiplying its positive-index coefficient by n^s and killing its constant coefficient. Equation (9.4) is equivalently

$$(1-w) \theta^s \mathfrak{F}(s, z, w) = z w \mathfrak{F}(s, z, w). \quad (9.10)$$

This is a coefficient-defined operator identity. No fractional-power branch for w is being assumed. Its spectral derivative uses the similarly diagonal multiplier $\log n$ on positive indices.

10 Polynomial truncations and elementary first spectral derivatives

Theorem 10.1 (Complete classification of polynomial truncations). *Fix $\operatorname{Re} s > 0$. The function $\mathfrak{F}(s, z, w)$ is a polynomial in w if and only if $z = 0$ or $z = -m^s$ for some positive integer m . In the latter case its degree is exactly m , and*

$$P_m^*(s, w) := \mathfrak{F}(s, -m^s, w) = 1 - \sum_{n=1}^m \left(\frac{m}{n}\right)^s \prod_{k=1}^{n-1} \left(1 - \left(\frac{m}{k}\right)^s\right) w^n. \quad (10.1)$$

For every integer $r \geq 1$ and $|w| < 1$,

$$\boxed{\sum_{d \geq 1} (-m)^d (\partial_s + d \log m)^r \operatorname{Li}_{\{s\}^d}(w, 1, \dots, 1) \Big|_{s=1}} = \partial_s^r P_m^*(s, w) \Big|_{s=1}. \quad (10.2)$$

The right side is a finite polynomial in w , with coefficients in the algebra over $\mathbb{Q}$ generated by $\log 2, \dots, \log m$.

Proof. At $z = -m^s$, the factor indexed by $k = m$ in (9.3) kills every b_n with $n > m$. The coefficient b_m is nonzero, since $k^s \neq m^s$ for $k < m$; their moduli are distinct. This proves (10.1) and its degree. Conversely, if a nonconstant sequence terminates, some b_n with $n \geq 2$ vanishes. Its product expression and $z \neq 0$ force $z = -k^s$ for one of the factors. The same modulus argument makes this k unique.

Along $z(s) = -m^s$, the derivative of the depth coefficient $z(s)^d$ acts as multiplication by $d \log m$. Local uniform convergence from Theorem 9.1 allows differentiation along this curve. This proves (10.2). At $s = 1$, derivatives of the finitely many positive-base powers in (10.1) have the asserted logarithmic coefficients. $\square$

The moving-depth derivative in (10.2) is not the fixed-depth spectral derivative. The distinction can be resolved exactly at first order. Put

$$T_m(w) = \partial_s \mathfrak{F}(s, -m, w) \Big|_{s=1}, \quad Q_m(w) = \partial_s P_m^*(s, w) \Big|_{s=1}. \quad (10.3)$$

Theorem 10.2 (Elementary first resonant derivative). *For every $m \geq 1$ and $|w| < 1$,*

$$\boxed{T_m(w) = Q_m(w) - m \log m (1 - w)^m \operatorname{Log}(1 - w)}. \quad (10.4)$$

Here Q_m is an explicitly computable polynomial of degree at most m . Equivalently,

$$T_m(w) = -(1 - w)^m \sum_{n=1}^m (-1)^n \binom{m}{n} n \log n \int_0^w \frac{t^{n-1}}{(1 - t)^m} dt. \quad (10.5)$$

All integrals in (10.5) are elementary. In particular,

$$\boxed{\sum_{d \geq 1} (-2)^d \partial_s \operatorname{Li}_{\{s\}^d}(w, 1, \dots, 1) \Big|_{s=1}} = -2 \log 2 \left[w(1 - w) + (1 - w)^2 \operatorname{Log}(1 - w) \right]. \quad (10.6)$$

Proof. Differentiate $P_m^*(s, w) = \mathfrak{F}(s, -m^s, w)$ at one. Since $\partial_z \mathfrak{F}(1, z, w) = -\operatorname{Log}(1 - w)(1 - w)^{-z}$, the chain rule gives $Q_m = T_m + m \log m (1 - w)^m \operatorname{Log}(1 - w)$, proving (10.4). This already proves elementary reducibility: Q_m is obtained by differentiating the finite expression (10.1).

For an independent constructive expression, differentiate (9.10) at $s = 1, z = -m$, where $\mathfrak{F} = (1 - w)^m$. The result, after division by $w(1 - w)$ and analytic extension at zero, is

$$T'_m(w) + \frac{m}{1-w} T_m(w) = - \sum_{n=1}^m (-1)^n \binom{m}{n} n \log n w^{n-1}, \quad T_m(0) = 0. \quad (10.7)$$

The integrating factor $(1 - w)^{-m}$ gives (10.5). Finally $P_2^*(s, w) = 1 - 2^s w + (2^s - 1)w^2$, so $Q_2 = -2 \log 2 w(1 - w)$; insert this into (10.4). $\square$

An explicit formula for the elementary integral is

$$\begin{aligned} \int_0^w \frac{t^{n-1}}{(1-t)^m} dt &= \sum_{\substack{0 \leq j \leq n-1 \\ j \neq m-1}} (-1)^j \binom{n-1}{j} \frac{1 - (1-w)^{j-m+1}}{j - m + 1} \\ &\quad + \mathbf{1}_{n=m} (-1)^m \text{Log}(1-w). \end{aligned} \quad (10.8)$$

Substituting this finite sum into (10.5) gives a direct rational-coefficient algorithm for Q_m . Two further examples are

$$Q_3(w) = -3w \log 3 + \left(\frac{15}{2} \log 3 - 3 \log 2 \right) w^2 + \left(3 \log 2 - \frac{9}{2} \log 3 \right) w^3, \quad (10.9)$$

$$\begin{aligned} Q_4(w) &= -8w \log 2 + 22w^2 \log 2 + \left(4 \log 3 - \frac{80}{3} \log 2 \right) w^3 \\ &\quad + \left(\frac{38}{3} \log 2 - 4 \log 3 \right) w^4. \end{aligned} \quad (10.10)$$

For $m = 1$, $\mathfrak{F}(s, -1, w) = 1 - w$ for every $\text{Re } s > 0$, so every positive spectral derivative is identically zero.

Corollary 10.3 (Elementary coefficient certificate). *The coefficients of T_m are*

$$[w^n]T_m = \begin{cases} (-1)^n \binom{m}{n} \left[-\log n + m \sum_{k=1}^{n-1} \frac{\log k}{k-m} \right], & 1 \leq n \leq m, \\ \frac{(-1)^m m \log m}{(n-m) \binom{n}{m}}, & n > m. \end{cases} \quad (10.11)$$

Proof. For $n \leq m$ differentiate the nonvanishing product in (9.3) logarithmically. For $n > m$, only the derivative of its zero factor $1 - m^{1-s}$ contributes. The remaining rational product at one is $(-1)^m m / ((n-m) \binom{n}{m})$, yielding the second line. $\square$

These formulas give a coefficientwise proof check without evaluating a multiple polylogarithm numerically. They also show why an infinite sum over depths can simplify even when its separate depth components look complicated.

11 All higher resonant jets and bounded nesting depth

The first derivative is not an isolated coincidence. Its zero-factor calculation yields a finite Bell formula at every spectral order. For $m, n \geq 1$ and $j \geq 1$ define

$$L_{j,n}^{(m)} = -\mathbf{1}_{j=1} \log n + (-1)^{j+1} \sum_{\substack{1 \leq k < n \\ k \neq m}} (\log k)^j \text{Li}_{1-j}(m/k). \quad (11.1)$$

The polylogarithms here have nonpositive integer orders, so they are *rational functions* of m/k ; no continuation across a logarithmic cut is involved. The excluded value $k = m$ is exactly the isolated zero factor. For example,

$$L_{1,n}^{(m)} = -\log n + m \sum_{\substack{k < n \\ k \neq m}} \frac{\log k}{k - m}, \quad (11.2)$$

$$L_{2,n}^{(m)} = -m \sum_{\substack{k < n \\ k \neq m}} \frac{k(\log k)^2}{(k - m)^2}. \quad (11.3)$$

Write $Y_r(L)$ for the Bell polynomial in $L_{1,n}^{(m)}, \dots, L_{r,n}^{(m)}$ with these m, n fixed.

Theorem 11.1 (Every fixed-weight spectral coefficient). *Let*

$$U_r^{(m)}(w) = \partial_s^r \mathfrak{F}(s, -m, w)|_{s=1} = \mathbf{1}_{r=0} + \sum_{n \geq 1} c_{r,n}^{(m)} w^n.$$

For every $r \geq 0$,

$$c_{r,n}^{(m)} = \begin{cases} (-1)^n \binom{m}{n} Y_r(L), & 1 \leq n \leq m, \\ \frac{(-1)^m m}{(n-m) \binom{n}{m}} \sum_{j=1}^r (-1)^{j+1} \binom{r}{j} (\log m)^j Y_{r-j}(L), & n > m. \end{cases} \quad (11.4)$$

The empty sum on the second line is zero. The coefficient series converges locally uniformly in $|w| < 1$ and represents the indicated analytic spectral derivative.

Proof. For $n \leq m$, none of the factors in $b_n(s, -m)$ vanishes at one. Choose the analytic logarithm of $b_n(s, -m)/b_n(1, -m)$ near one. The derivatives of a nonzero factor are

$$\partial_s^j \log(1 - mk^{-s}) \Big|_{s=1} = (-1)^{j+1} (\log k)^j \text{Li}_{1-j}(m/k), \quad k \neq m. \quad (11.5)$$

For $|mk^{-s}| < 1$ this follows from the logarithmic series. For the other finitely many k , it follows by differentiating the rational first logarithmic derivative; equivalently both sides are the same rational function of mk^{-s} . The term n^{-s} contributes only $-\log n$ to the first logarithmic derivative. Thus all logarithmic derivatives are exactly (11.1). Exponentiating their Taylor series with (2.7) proves the first line of (11.4).

For $n > m$, factor the coefficient as

$$b_n(s, -m) = (1 - m^{1-s}) R_n(s), \quad R_n(s) = -mn^{-s} \prod_{\substack{k < n \\ k \neq m}} (1 - mk^{-s}). \quad (11.6)$$

Now $R_n(1) = (-1)^m m / ((n-m) \binom{n}{m})$ and its logarithmic derivatives are again $L_{j,n}^{(m)}$. Also

$$\partial_s^j (1 - m^{1-s}) \Big|_{s=1} = (-1)^{j+1} (\log m)^j \quad (j \geq 1),$$

while its value is zero. Leibniz's rule and the Bell expansion for R_n give the second line. Local uniform convergence and identification with $U_r^{(m)}$ follow from Theorem 9.1. $\square$

The first three tail polynomials multiplying $R_n(1)$ are

$$B_1 = \ell, \quad (11.7)$$

$$B_2 = 2\ell L_1 - \ell^2, \quad (11.8)$$

$$B_3 = 3\ell(L_1^2 + L_2) - 3\ell^2 L_1 + \ell^3, \quad (11.9)$$

where $\ell = \log m$. The vanishing factor lowers the number of harmonic sums that can occur by one. This is the mechanism behind the depth bound.

Definition 11.2 (Comparison class for the depth reduction). A shifted logarithmic Euler sum of nesting depth $d + 1$ is a convergent sum of the form

$$\sum_{n > k_1 > \dots > k_d \geq 1} w^n R_0(n) (\log n)^{p_0} \prod_{i=1}^d R_i(k_i) (\log k_i)^{p_i}, \quad (11.10)$$

with finitely many stated index exclusions, nonnegative integers p_i , and rational functions R_i with rational coefficients. In the present application the exclusions are $n > m$ and $k_i \neq m$, and the poles are integer shifts arising from $0, 1, \dots, m$. Finite polynomials are included as depth-zero terms.

Corollary 11.3 (All-depth to bounded-nesting reduction). *For fixed $m \geq 1$ and $r \geq 1$, the function*

$$\sum_{d \geq 1} (-m)^d \partial_s^r \text{Li}_{\{s\}^d}(w, 1, \dots, 1) \Big|_{s=1} \quad (11.11)$$

is a finite linear combination of a polynomial and shifted logarithmic Euler sums of nesting depth at most r , with coefficients in $\mathbb{Q}[\log m]$. The coefficients and summands are explicitly generated by (11.4). This is a representation theorem in the comparison class (11.10), not a minimal-depth or arithmetic independence theorem for ordinary multiple polylogarithms.

Proof. Every Y_{r-j} on the second line of (11.4) is a polynomial with at most $r - j \leq r - 1$ factors of the single harmonic sums in (11.1); factors $\log n$ do not introduce another summation index. Expand their products by partitioning the index tuples into strict order patterns and equality blocks. An equality block multiplies rational kernels and adds logarithmic exponents, staying in the stated class. This is the finite stuffle decomposition and produces at most $r - 1$ inner nested indices. The outer index n raises the total nesting depth to at most r . Rationality of every kernel follows from the nonpositive polylogarithm orders in (11.1). Absolute convergence for $|w| < 1$ justifies each finite rearrangement. $\square$

Example 11.4 (The second derivative at depth weight -2). Let $\ell = \log 2$ and

$$M_n = 2 \sum_{k=3}^{n-1} \frac{\log k}{k-2} - \log n.$$

The theorem gives the explicit depth-two representation

$$U_2^{(2)}(w) = \ell^2 w^2 + 4 \sum_{n=3}^{\infty} \frac{w^n (2\ell M_n - \ell^2)}{n(n-1)(n-2)}. \quad (11.12)$$

This is an exact identity, not a claim that the second derivative is elementary. The first derivative's elementary form does not imply the same conclusion at higher orders.

Corollary 11.5 (Convergent endpoint cancellation). *For every $m \geq 1$ and $r \geq 0$, the coefficient series for $U_r^{(m)}$ converges absolutely at $w = 1$, and*

$$U_r^{(m)}(1) = 0. \quad (11.13)$$

Here the endpoint is defined by the coefficient series in n , or equivalently its Abel limit. It is not a termwise substitution of $w = 1$ into the generally divergent individual-depth multiple polylogarithms.

Proof. For large k , $\text{Li}_{1-j}(m/k) = O(1/k)$, so $L_{j,n}^{(m)} = O(1 + \log^{j+1} n)$ at fixed m, j . Bell monomials of weight $r - j$ grow at most as a fixed multiple of $1 + \log^{2(r-j)} n$. Since $R_n(1) = O(n^{-m-1})$, the tail in (11.4) is absolutely summable; the case $r = 0$ is a polynomial.

For every finite N , the coefficient sum through $n = N$ is the r -th spectral derivative at one of $P_N(s, -m; 1)$ (including its constant for $r = 0$). Write this product as

$$P_N(s, -m; 1) = e^{-mC_N(s;1)} \mathcal{G}_N(s, -m; 1).$$

The derivatives of $\mathcal{G}_N$ of each fixed order at one are bounded uniformly in N by Theorem 3.1. The exponential contributes $(N + 1)^{-m}$ times a polynomial in $\log(N + 1)$ of bounded degree when differentiated a fixed number of times, by (3.2). Their product therefore tends to zero. Absolute convergence identifies this limit with (11.13). $\square$

For instance (11.12) yields the convergent identity

$$\sum_{n=3}^{\infty} \frac{2 \sum_{k=3}^{n-1} \log k / (k - 2) - \log n}{n(n-1)(n-2)} = 0. \quad (11.14)$$

This low-order example also follows by interchanging its two absolutely convergent sums and telescoping the rational tail. The all-order theorem explains the entire Bell family of such cancellations.

12 Verification, source audit, and integration scope

12.1 Reproducible evidence and its limits

The analytic theorems are justified by the proofs above. The delivered scripts are implementation checks and transparent finite certificates. They do not turn an approximate transcendental equality into a proof. The actual executed suites are summarized in Table 1.

The exact checker uses SymPy 1.14.0 and exact rational arithmetic. In the first-resonance coefficient comparison, the symbols for $\log 2, \log 3, \dots$ are kept formally independent; the check does not rely on numerical approximations or unproved independence of the actual logarithms. The finite Laurent arrays are generated by Newton's recurrence, separately with and without the spectral pole, before their coefficient conversion is compared. This tests in particular both factorials in (5.4).

The numerical suites use Python 3.13.5 and mpmath 1.3.0. The largest scaled residual among the 99 whole-function comparisons was approximately 3.35×10^{-50} ; the largest absolute residual was approximately 1.13×10^{-48} . The higher-jet and cyclotomic suites had largest scaled residuals approximately 7.16×10^{-60} and 1.02×10^{-60} , respectively. The JSON outputs retain the individual parameters, acceptance thresholds, and observed residuals. These figures describe the executed arithmetic, not a certified error interval.

Suite	Checks	Executed scope
Exact algebra	826	Newton products; diagonal Laurent conversion through depth four and Laurent index two; first resonances $1 \leq m \leq 12$, $1 \leq n \leq 48$; rational negative-order polylogs through order eight.
Whole-function numerics	99	Gamma product, shifts, multiplication, Stieltjes-zeta jets, polylog primitives, zero values, first resonances, moving truncations, and Cauchy finite-part extraction.
Higher-jet coefficients	960	Two independent coefficient constructions for $1 \leq m \leq 6$, $1 \leq n \leq 32$, and spectral orders zero through four.
Cyclotomic products	16	Genera two through five, two positive Hurwitz parameters, and resonant and nearby complex spectral points.

Table 1: Executed verification. The last three suites use 60 decimal-digit floating-point arithmetic; they are not interval certificates.

Independent representations are used where practical. Gamma ratios are compared with finite centered products and Hurwitz-zeta tails. Spectral zeta series are compared with finite polylogarithmic heads and separately accelerated tails. First resonant derivatives are obtained by propagating the differentiated coefficient recurrence and compared with a derivative of the finite moving-root polynomial. Higher coefficients compare that recurrence with the zero-factor Bell formula. Diagonal finite parts are also extracted by a finite-radius Cauchy average of the raw Newton polynomial, rather than obtained only by subtracting two truncated formal expressions.

For a reproducible numerical product evaluation one may split at an integer N and use, for example,

$$\begin{aligned} \mathcal{G}(s, z; a) = & e^{z(\zeta(s, a) - 1/(s-1))} \prod_{n < N} (1 + zx_n^{-s}) e^{-zx_n^{-s}} \\ & \times \exp \left(\sum_{k \geq 2} \frac{(-1)^{k-1} z^k}{k} \zeta(ks, N + a) \right). \end{aligned} \quad (12.1)$$

The tail is convergent when $|z| < (N + a)^{\operatorname{Re} s}$. The supplied code truncates this convergent tail and checks consistency with independent formulas; it does not claim to enclose every floating-point rounding error. At $s = 1$, the removable linear term is evaluated directly as $-\psi(a)$.

12.2 A confirmed editorial issue, already noticed in incoming work

In the inspected integration chapter, the historical discussion following the low-order log-Gamma integrals says: “Basis atoms must be $\mathbb{Q}$ -independent.” Taken literally, this is too strong. One often cannot prove independence of the retained numerical periods. A valid search protocol can still remove *known* dependencies and require a proposed relation to involve its target, then verify or prove it independently. A suitable replacement is:

Remove proved dependencies from the comparison basket, or work in an explicitly specified quotient. Independence of the retained evaluated constants need not be known.

A returned relation must be checked for its target coefficient and then subjected to an independent proof or a clearly labeled numerical validation.

The incoming shifted-Hurwitz and Stieltjes-calculus reports already propose a correction of this kind. We confirm the source wording at the pinned revision and retain the correction as an integration note, not as a newly discovered flaw. The current pointwise harmonic-number Stieltjes derivative ladder is not contradicted by anything in this article.

12.3 Extension guards established by this paper

Several tempting extensions would be false or underspecified. They are listed here as guards for integrating *our* formulas, not allegations that the upstream manuscript made these assertions.

First, $g_d(a)$ is not the diagonal Laurent finite part: already (5.6) exhibits the extra $-\gamma_1(a)$. Second, parameter multiplication needs the exponential in (6.2); the Gamma slice is an immediate check. Third, differentiating a moving depth weight is not fixed-weight spectral differentiation; (10.4) records the missing logarithmic term. Fourth, a zero of $\mathcal{G}(1, z)$ cannot be substituted naively into a meromorphic logarithmic derivative: the finite value (4.9) results from cancellation.

Fifth, the all-depth convergence domain cannot simply include $s = 0$, as (9.9) shows. Sixth, the nesting bound in Corollary 11.3 refers to shifted logarithmic Euler sums, not to a proof of minimal depth or numerical independence in a smaller ordinary polylogarithm algebra. Finally, the endpoint identity (11.13) uses the convergent largest-index representation, not a sum of undefined individual all-ones multiple-zeta values.

12.4 Repository placement and unresolved claims

The package contains a standalone source, compiled PDF, verification code, recorded output, a provenance manifest, and integration notes. The incoming README treats archives as intake material until they are placed and written into a thematic report; preparation of this delivery does not perform that intake. No upstream file has been changed.

A natural placement is a new report under the polylogarithm programme, with cross-references from the integration and differentiation chapters to Sections 3–8, and from the depth chapter to Sections 9–11. The target chapter’s numbering, theorem labels, and notation should be reconciled on intake; the proposed prefix for imported labels is `nhj`.

The source already contains an exact S_4 proof. This report neither reproves it as a new result nor resolves the surviving short S_6 or newer S_8 candidates. It does not establish new transcendence or numerical period-independence statements, or claim that all displayed reductions are first in the literature. Its mathematical contribution is the proved families above, with their specified regularizations and comparison classes.

13 Further research questions

The identities suggest the following concrete continuations. These are questions for further work, not additional assertions of this article.

1. Smaller function classes for higher fixed-weight jets. Determine whether $U_2^{(m)}$ admits a finite expression using ordinary polylogarithms, their spectral derivatives, and elementary functions,

or whether a genuinely shifted two-index function is needed in a specified functional algebra. Equation (11.12) is an explicit first test. An obstruction would need a precise differential or symbolic framework; failure of a numerical search would not establish it.

2. Constructive conversion of the comparison class. Give an algorithm converting the rationally shifted logarithmic Euler sums in (11.10) into a canonical iterated-integral or Lerch-derivative vocabulary. Track separately the nesting depth, spectral derivative order, and the finite shift set. Such a result would sharpen Corollary 11.3 without making an arithmetic minimality claim.

3. Non-diagonal spectral directions. Replace the common exponent by several independent perturbation directions. Which symmetrized mixed jets still admit a finite-product model, and what is the correct analogue of the Bell zero-factor theorem? A useful first case is a two-colour periodic assignment of exponents.

4. Path dependence of multiple Laurent finite parts. For a genuinely multivariable nested Hurwitz zeta function, compare limits along $s_j = 1 + c_j \varepsilon$ with the equal-direction formula (5.4). Identify the dependence on the direction vector and the additional algebraic conditions needed for a shuffle- or stuffle-compatible renormalization. This question must be compared carefully with the existing renormalization literature, rather than treated as unexplored in general.

5. Individual higher-genus reflection formulas. The cyclotomic norm (7.4) is explicit, but it does not recover each individual $\mathcal{G}^{[m]}(1/m, z; a)$. Find connection or reflection laws that complement this norm, particularly at rational a and for $m = 2, 3$. Specify growth and logarithm normalizations before proposing uniqueness of a factorization.

6. Mixed conductor and cyclotomic reductions. Study which combinations of rational parameter multiplication and rotation in the depth variable reduce the individual fractional-order spectral jets to a finite Dirichlet- L -derivative vocabulary. Equation (7.5) settles one trace, not all character components.

7. Higher spectral values at moving Gamma zeros. Construct an explicit finite algorithm for $F_r(-a - m; a)$ for every r , using the isolated zero factor and pole-free logarithmic jets of the remaining product. Determine the smallest natural Stieltjes–zeta coordinate set needed for those evaluations. Proposition 4.2 gives exact compatibility constraints for any proposed formulas.

8. Argument continuation and monodromy. Continue $U_r^{(m)}(w)$ from the disk across arcs away from $w = 1$ and give an exact monodromy calculus. The first-order formula is elementary, but the higher shifted sums may require a larger alphabet. The coefficient recurrence supplies a normalized local solution from which to start.

9. Differentiated endpoint identities. Determine the exact finite parts after applying powers of $w\partial_w$ to the convergent endpoint sums in Corollary 11.5. At the first nonsummable power, identify the subtraction polynomial explicitly. This is an identity problem about normalization, not merely a tail bound.

10. Primitive evaluations at arithmetic depth weights. Use (8.3) and (6.5) to isolate combinations of ordinary and Euler primitives at rational or algebraic z/a . Which such combinations reduce to Gamma, polygamma, Dirichlet- L jets, and a finite set of algebraic-argument polylogarithms? Any claimed reduction should retain the underlying branch domain.

11. Connection with the remaining Gaussian mixed candidates. Investigate deformations with colour and unequal first order that specialize to the repository's S_6 or S_8 comparison baskets. The equal-order resonance theorem does not currently provide those reductions. A concrete next step is to derive the coloured finite-product or recurrence identity before making a new relation search.

12. Formal and certified interfaces. Formalize Newton extraction, the regulator-conversion recurrence, the cyclotomic finite-cutoff identity, and the Bell zero-factor theorem before attempting the analytic limits. Separately, add interval certification to the numerical tail evaluators, including cancellation near reciprocal-Gamma zeros. Neither project should be conflated with the currently executed floating-point checks.

Conclusion

The equal-order nested family has more exact structure than its individual depth components suggest. A finite product carries the harmonic coordinates, its pole subtraction produces a Gamma-valued holomorphic family, and its Abel transform produces the all-depth multiple polylogarithm. Spectral jets therefore have two effective normal forms: Stieltjes-zeta Bell coefficients at the singular spectral slice, and logarithmic harmonic Bell coefficients at a resonant depth weight. The cyclotomic identity supplies a further exact bridge from fractional spectral orders to ordinary Gamma.

The main cautions are mathematical rather than merely editorial. A diagonal Laurent finite part is not a harmonic cutoff constant, a moving depth weight is not a fixed parameter, and bounded nesting in a stated shifted-sum class is not numerical period independence. Keeping those distinctions explicit allows the new formulas to be integrated without reviving the normalization and evidence problems that the source manuscript itself works to avoid.

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